

Multi-Source Telemetry Console

Preserve health evidence across presentation and records

A deterministic, receive-only project with no actuator

Lesson	027 — project	Time	deterministic 18-second fixture
Prerequisites	Lessons 001–026	Board	Arduino Mega 2560
Evidence	RGB cadence and D13	Status	host verification; bench open
Energy class	E1: USB, passive buttons, resistor-limited LEDs		

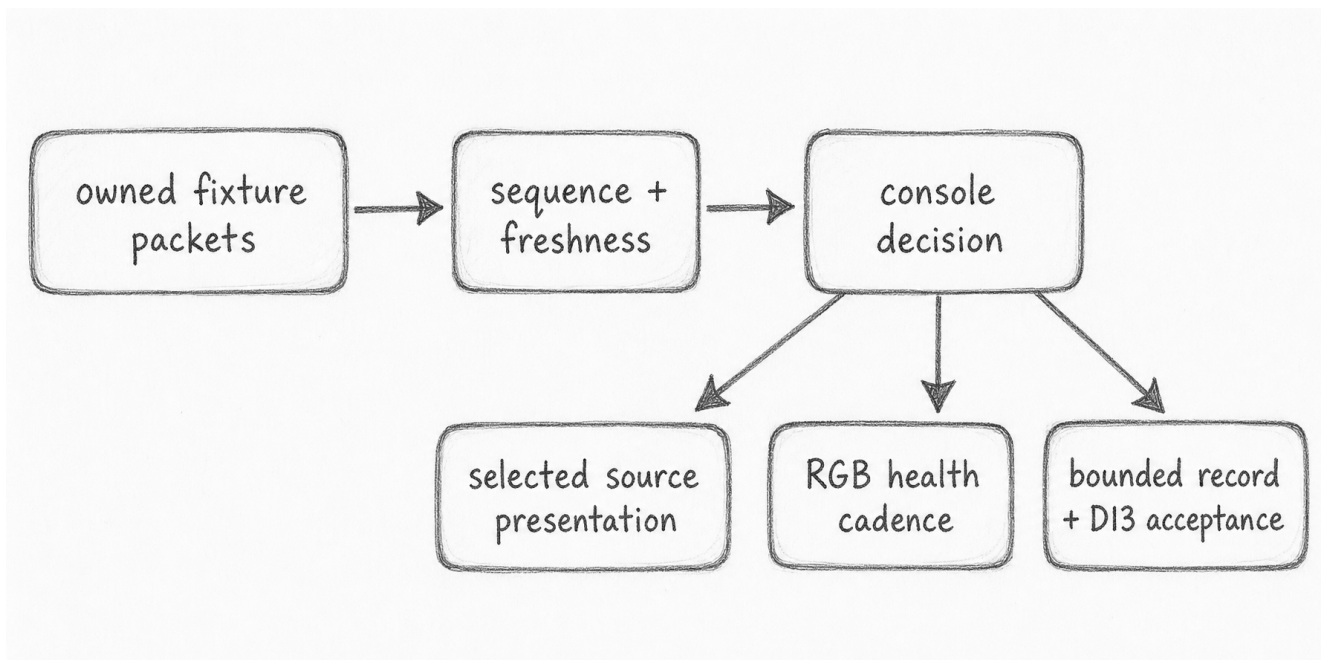
Boundary. This is an educational annunciator for owned lab fixtures. It is not a fire, burglar, medical, industrial, environmental-safety, or emergency system. It does not actuate anything, dispatch help, transmit, replay, clone, or interpret an unknown protocol.

1 Mission and evidence chain

Compose the receive-only arc without losing the distinction between evidence and presentation:

source evidence → normalized reading → console decision → visible state → stable record.

The released exercise uses three compiled-in, owned fixture observations: temperature, relative humidity, and contact. A live receiver is not required. Input order cannot change configured source order. Missing sources remain explicit. Presentation and record failures cannot rewrite health.

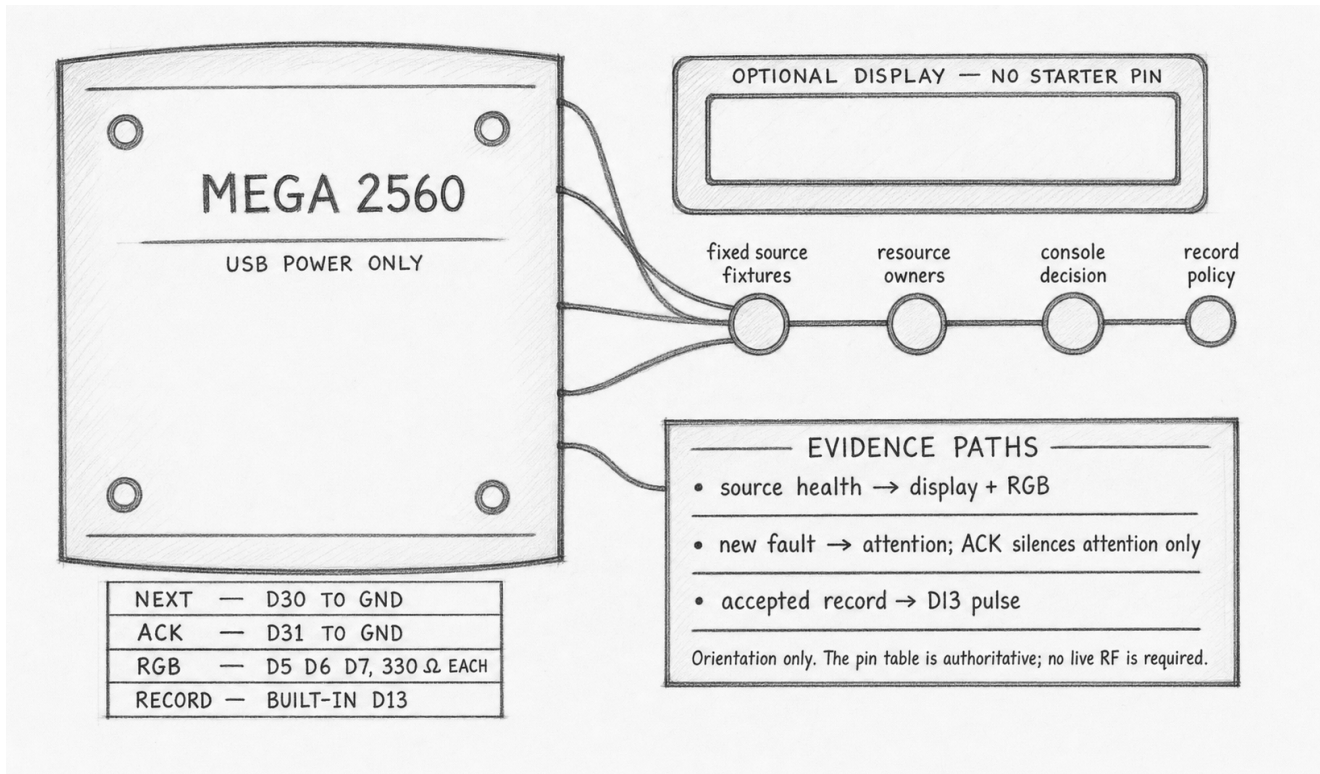


1.1 Learning outcomes

- distinguish capture, frame, packet, record, and presentation;
- explain configured slot order and explicit missing observations;

- derive health from quality, sequence condition, and freshness;
- keep acknowledgement separate from health;
- prove record acceptance without confusing it with source validity;
- replay an exact mixed-source trace with byte-identical output.

2 Orientation and circuit plan



Alternative text: graphite orientation plate showing the USB-powered Mega, optional unassigned display, D30 next and D31 acknowledge buttons to ground, D5–D7 RGB channels with one 330-ohm resistor each, built-in D13 record evidence, and separate health and record paths. No live radio or actuator is shown.

The plate shows the extension path as well as the starter circuit. The released sketch's pin table below is authoritative; the display block is optional further work and owns no pin in the starter circuit.

Circuit element	Released role	Electrical constraint
Next button, D30	advance configured source slot	internal pull-up; switch to GND
Acknowledge button, D31	acknowledge current attention intent	internal pull-up; switch to GND
RGB LED, D5–D7	health plus non-color cadence	one 330 ohm resistor per channel
D13 LED	toggles after complete record acceptance	built-in LED, independent of health
Source fixtures	deterministic observation evidence	compiled-in, owned lab values

The runtime owns every endpoint. The project borrows the console, buttons, presentation, and record sink; each borrower has a shorter lifetime than its owner. The starter circuit has one resistor per RGB channel, common ground, and no display, sounder, relay, external load supply, transmitter, or mains connection.

3 Vocabulary

Word	Meaning in this arc	What it does not mean
Capture	timestamped physical evidence	decoded command
Frame	bounded protocol candidate	trusted packet
Packet	ADK-owned telemetry envelope	authenticated identity
Record	stable storage representation	source truth
Presentation	human-visible interpretation	mutation of observation health

CRC detects accidental corruption. It does not authenticate a sender, hide content, prove a location, or prevent replay. Source identifiers organize owned fixtures; they are not credentials. Freshness bounds local age; it does not prove a remote clock.

4 The narrative loop

```
observeTelemetrySources (now);
decideConsoleState      (now);
presentConsoleState     ();
recordConsoleEvidence   ();
```

observe builds the fixed observation fixture. **decide** submits one complete configured-slot view through **TelemetryConsoleProject**. **present** checks the RGB presentation result. **record** checks the bounded record result. No stage hides time or allocates memory. The sketch uses wrap-safe elapsed time and shuts down after 18 seconds regardless of button input. Shutdown turns off both evidence paths and releases their pins.

5 Health precedence

Health	Required interpretation	Evidence
Starting	expected slots have not all produced usable observations	blue cadence
Healthy	every expected slot is present, valid quality, and fresh	green cadence
Degraded	an observation is aging or reports a sequence gap	amber cadence
Fault	stale, invalid, corrupt, duplicate, or missing evidence	fast red cadence
Stopped	lifecycle inactive	unlit

Fault dominates degraded; degraded dominates healthy; startup is not silently healthy. When several conditions coexist, the same input snapshot must always choose the same result.

6 Operator intent is not evidence

The next action selects a configured slot, not a source ID or packet-arrival position. A held button produces one debounced event. Selection wraps after the last configured slot.

Acknowledgement applies only to the current attention intent. The starter circuit has no sounder; inspect the state in host replay, then add a bounded sounder only as the extension exercise. Acknowledgement does not:

- convert fault to healthy;
- clear a missing or stale source;
- erase or rewrite a record;
- prevent a new fault from being announced.

The automated replay holds a fault across acknowledgement: fast red evidence remains while **Attention** becomes **Notice**. Recovery followed by a different fault produces **Attention** again.

7 Presentation and record separation

The released presentation converts health to an RGB color and a distinct cadence. Its selected-source argument stays available for a later display without entering the decision. A presentation failure leaves the source snapshot unchanged.

The record policy writes on accepted observation, health transition, acknowledgement, or periodic heartbeat. Simultaneous reasons become one record under fixed priority. A failed write remains visible and may retry only under a bounded deterministic policy. No queue grows in SRAM.

Failure	Must remain unchanged	Separate evidence
presentation write	source health and pending record	returned status
record append	source health and selected slot	D13 does not toggle
record retry	accepted observation snapshot	same bounded record bytes
attention extension	health and record intent	red evidence remains

8 Deterministic fixture progression

The canonical sketch supplies this fixed, dependent progression. Each row retains every source introduced by the preceding row:

Elapsed time	Source set	Expected health
0–2 s	all configured sources missing	Starting
2–3 s	temperature present, startup grace active	Starting
3–4 s	temperature present, startup grace expired	Fault
4–6 s	temperature and humidity present	Fault
6–9 s	all three sources fresh	Healthy
9–12 s	humidity aging	Degraded
12–15 s	humidity stale	Fault
15–18 s	all three sources fresh again	Healthy
18 s	fixed shutdown, independent of button input	Stopped

The host suite separately replays identical timestamps, packet bytes, and button events twice and compares health, signal intent, selected slot, record reasons, and record bytes.

9 Predict, observe, interpret

9.1 Resource acquisition

Predict: blue cadence appears only after all required owners initialize. **Observe:** startup cadence without Serial. **Interpret:** this supports resource acquisition; it does not prove shutdown release or a valid source.

9.2 Source health

Predict: the trace advances blue, green, amber, red. **Observe:** the RGB cadence at TP-R, TP-G, and TP-B. **Interpret:** agreement supports the console decision. It does not authenticate a fixture or establish sensor accuracy.

9.3 Record acceptance

Predict: D13 toggles only after a complete record is accepted. **Observe:** inject a failed append in the host fixture and compare the acceptance state with health evidence. **Interpret:** D13 proves the simulated storage call path, not physical media durability.

9.4 Safe shutdown

Predict: RGB and record evidence become inactive and owned pins release. **Observe:** measure D5–D7 and D13 against GND after the stop path. **Interpret:** inactive LEDs alone do not prove high impedance or claim release.

10 Diagnosis

Observation	Check in order
stays Starting	configured count, explicit presence, first observation time
Fault instead of Degraded	stale boundary, invalid quality, duplicate ID, missing slot
selection follows arrival	configured-slot table; never sort by packet arrival
acknowledgement clears red	separate sound acknowledgement from health state
D13 toggles on failed record	acceptance must follow a successful append
new fault lacks attention	fault-episode transition and acknowledgement reset
different second replay	hidden clock, mutable fixture, uninitialized padding, ordering

11 Automated verification

```
make host-test
make host-test-sanitize
make arduino-Lesson027TelemetryConsole
make size-check
make doc/lessons/027.pdf
make lessons-check
make site-check
```

These targets verify code and publication. They do not establish physical verification. There is intentionally no transmit, replay, clone, brute-force, scan, or protocol-export target.

Package status

The interface, deterministic host tests, canonical Mega sketch, and printable package are present. Physical Mega acceptance remains open in the repository's separate acceptance campaign. The exercise makes no live-radio or multi-room hardware claim.